Executive Summary

This analysis examines the relationship between price positioning, customer engagement, perceived quality, and brand strategy within a large beauty product portfolio. The goal was not simply to describe performance, but to understand what truly drives customer attention versus long-term satisfaction, and where strategic risks or growth opportunities exist.

Business Context

Beauty portfolios often rely on metrics like popularity, ratings, and promotional tactics to guide decisions. However, these signals can conflict. Products that attract attention may not deliver the strongest customer satisfaction, while highly rated products may remain under-leveraged. The core question guiding this study was:

> Is portfolio performance driven more by value perception an brand strength than by pricing or promotional tactics?

Approach

Using product-level data covering engagement metrics (love_count), ratings, pricing, brand attributes, and merchandising signals, the analysis evaluated portfolio performance across categories, price tiers, discounts, marketing labels, and brands. The goal was to identify structural patters rather than isolated product wins.

Headline Findings

1. Engagement is heavily concentrated in budget-priced products.
   Lower-priced products account for the majority of customer interaction and portfolio visibility, making affordability the primary driver of engagement.

2. Higher price does not meaningfully improve perceived quality
   Premium products receive only slightly higher ratings despite large price jumps, suggesting limited differentiation in perceived value.

3. Popularity does not equal satisfaction
   Multiple highly loved products carry below-average ratings, revealing an "engagement-satisfaction gap" driven by hype, brand pull, or visibility rather than product quality.

4. Category rules are structurally different.
   - Makeup functions as the engagement engine of the portfolio
   - Skincare shows a stronger customer satisfaction but lower engagement intensity

   This indicates a balance opportunity between visibility and quality-led growth.

5. Promotions and marketing badges are weak performance drivers
   Discounting correlates with lower ratings and lower engagement, suggesting it is used as a corrective mechanism rather than a growth strategy. Among marketing attributes, only exclusivity shows consistent positive impact.

6. Brand identity outperforms pricing strategy

Several affordable brands generate engagement comparable to, or higher than, premium brands, indicating that brand positioning and perceived value matter more than price level alone.

Core Strategic Insight

Across the portfolio, customer attention is driven primarily by accessibility and brand strength, not by premium pricing or promotional tactics. At the same time, high engagement does not reliably signal strong customer satisfaction, creating a potential long-term risk if decision-making relies only on popularity metrics.

Strategic Implication

The portfolio appears structurally dependent on value-driven products for traffic and engagement, while premium and highly rated segments play smaller, positioning-focused roles. This creates both opportunity and risk:

- Opportunity to scale high-satisfaction categories and brands that are currently underexposed.
- Risk of over-reliance on hype-driven engagement that may erode long-term trust.

Business Problem & Objective

Business Problem

In large beauty portfolios, performance is often evaluated using surface-level metrics such as popularity, ratings, pricing, and promotional activity. However, these signals so not always align. Products that attract high engagement may not deliver strong customer satisfaction, while highly rated products may fail to generate significant attention.

This creates a strategic challenge:

- Are customers engaging because products deliver real value, or because of hype, brand pull, or affordability?
- Is pricing aligned with perceived quality?
- Do promotional tactics like discounts and marketing labels actually drive performance?

Without understanding these relationships, portfolio decisions risk being driven by misleading indicators, leading to inefficient pricing, merchandising, and brand strategies.

Objective of the Analysis

The objective of this project is to evaluate what truly drives product performance across the portfolio by analysing the relationship between:

- Customer engagement (loves_count)
- Customer satisfaction (ratings)
- Price positioning
- Brand performance
- Category dynamics
- Promotional and marketing attributes

Specifically, the analysis aims to:

1. Identify where engagement and satisfaction align, and where they diverge.
2. Understand how pricing impacts perceived value and customer interaction.
3. Evaluate whether discounts and marketing attributes improve performance.
4. Assess brand efficiency relative to price positioning.
5. Detect structural dependencies within the portfolio that may create growth opportunities or strategic risks.

Expected Business Value

This analysis is designed to support strategic decision-making by helping stakeholders:

- Optimize portfolio composition across price tiers and categories
- Prioritize brands and products with sustainable performance signals.
- Reduce reliance on misleading engagement metrics
- Improve pricing, positioning, and merchandising strategies based on customer behaviour patterns rather than assumptions.

Dataset & Methodology

Dataset Overview

The analysis uses a product-level dataset from an e-commerce beauty platform, containing information on product attributes, customer engagement, pricing, merchandising labels, and category structure.

Each row represents one product, allowing portfolio-wide comparison across brands, categories, and price-tiers.

Key data groups included:

- Product information: product name, brand, category hierarchy
- Customer signals: Customer engagement (favorites/saves), rating, number of reviews
- Pricing data: regular price, sale price, displayed value price.
- Merchandising attributes: Indicators such as new launch, limited edition, online-only availability, exclusivity, and stock status.
- Product variation metadata: variation and size details (used selectively)

The dataset enabled analysis across oth performance metrics (engagement and satisfaction) and business levers (pricing and positioning).

Data Preparation & Cleaning

To ensure analytical reliability, the dataset underwent structured preprocessing:

1. Column Selection
   Fields with limited analytical value or inconsistent structure were excluded, including ingredient lists, variation descriptions, inconsistent size units, and noisy category levels. Non-essential identifiers were also removed to streamline analysis.

2. Data Quality Standardization
   Duplicate records were removed, and data types were standardized across numerical and categorical fields. Additional validation checks ensured that each record represented a single unique product.

3. Missing Data Treatment
   Missing values were handled using business-driven rules:
   - Engagement-related fields (favourites and review counts) were set to zero when absent
   - Ratings were intentionally left null to preserve the signal of products with no customer feedback
   - Missing category values were grouped under "Unknown"
   - Binary merchandising attributes were defaulted to zero, reflecting the absence of a badge or label

4. Derived Metrics
   Additional variables were created to support pricing and performance analysis:
   - Final Price: sale price where available, otherwise regular price
   - Discount Indicator: identifies discounted products
   - Discount Percentage: relative reduction from regular price

5. Segmentation Framework
   Products were grouped into standardized analytical segments to enable consistent comparison:
   - Price Segment: Budget, Mid, Premium
   - Rating Bucket: High Rated, Low Rated, No Rating
   - Discount Bucket: No, Low, Medium, High Discount

6. Data Validation & Outlier Handling
   Obvious pricing anomalies (e.g., zero or unrealistic prices) were removed. Validation checks were applied to identify negative pricing issues, and text fields were standardized through trimming and cleaning.

Analytical Methodology

The analysis followed a hypothesis-driven framework, designed to move from descriptive patterns to strategic insights.

Key evaluation dimensions included:

- Category performance (engagement vs satisfaction)
- Pricing and perceived value
- Discount impact
- Brand positioning effectiveness
- Portfolio composition across price tiers
- Engagement-satisfaction mismatch

Performance was assessed using:

- Average and total engagement (loves_count)
- Average ratings
- Product counts (to control for portfolio size effects)
- Coss-segment comparisons

The focus was not on predictive modelling but on strategic pattern detection, enabling business interpretation rather than purely statistical reporting.

Analytical Philosophy

A core principle of this analysis was to separate:

- Attention metrics (what attracts customers)
  From
- Quality signals (what satisfies customers).

The distinction allowed deeper understanding of portfolio dynamics and prevented engagement metrics from being treated as a direct proxy for product success.

Key Findings

This analysis reveals clear structural patterns in how customers engage with the portfolio. Rather than isolated observations, the findings show a consistent story: **customer attention is driven primarily by affordability and brand pull, while satisfaction follows a different logic.**

1. **Engagement and Satisfaction Follow Different Patterns**
   Across the portfolio, high engagement does not consistently align with high customer ratings.

- Several products with very high engagement show below-average ratings.
- Categories and brands attracting the most attention are not always the highest rated

This indicates an engagement-satisfaction gap, where visibility, trend relevance, or brand strength drive attention more strongly than product performance itself.

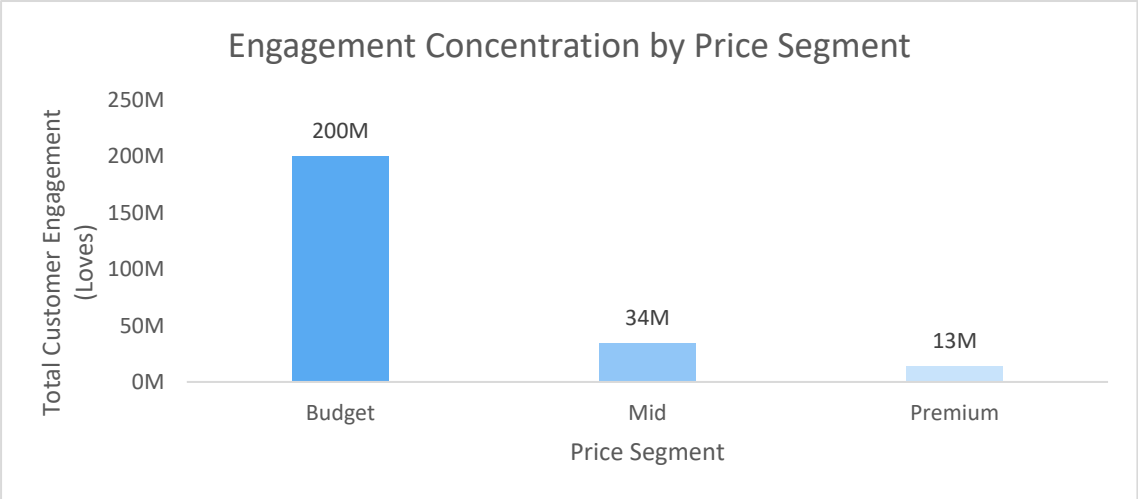
Implication: Engagement metrics alone are not reliable indicators of product quality or long-term customer satisfaction.

2. **Budget Products Drive Portfolio Engagement**
   Customer interaction is overwhelmingly concentrated in lower-priced products.

- Budget products represent the majority of the assortment.
- They generate a disproportionately large share of total engagement
- Engagement declines steadily as price increases.

This suggests that affordability is the primary driver of portfolio traffic an product discovery.

Implication: The portfolio's growth and visibility are heavily dependent on value- driven offerings.

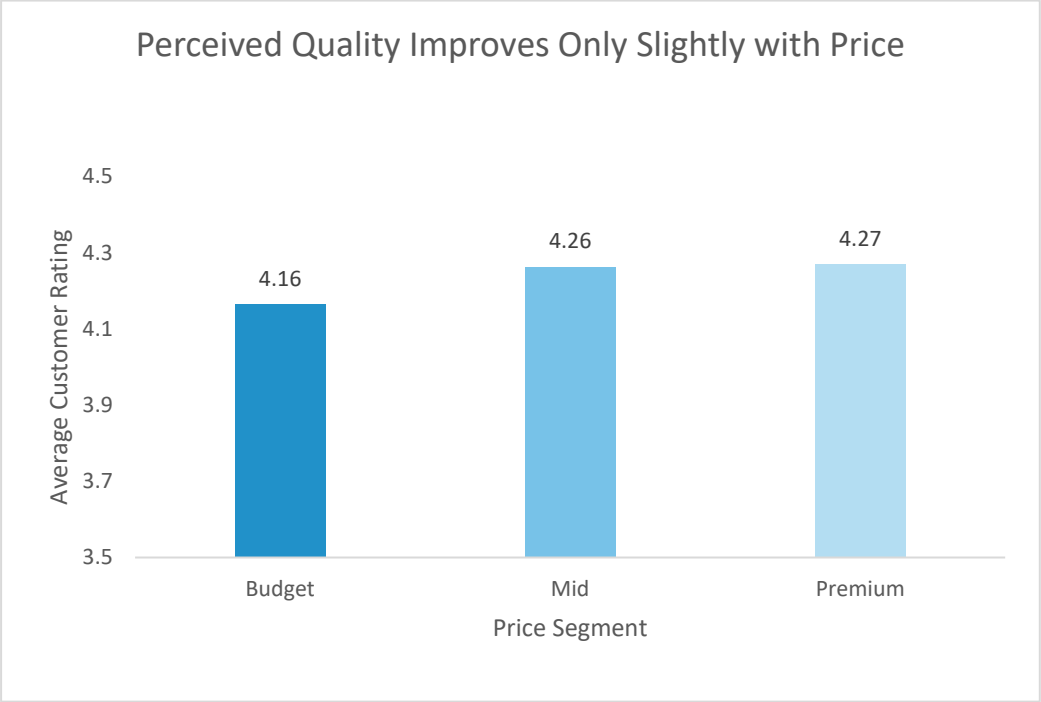


3. Premium Pricing Shows Limited Perceived Quality Advantage

While higher-priced products receive slightly higher ratings on average, the difference is marginal relative to the price increase.

- Rating improvement from budget to premium tiers is small
- Premium positioning does not translate into significantly stronger customer satisfaction

Implication: Price alone is not creating clear perceived quality differentiation, raising questions about premium value communication.



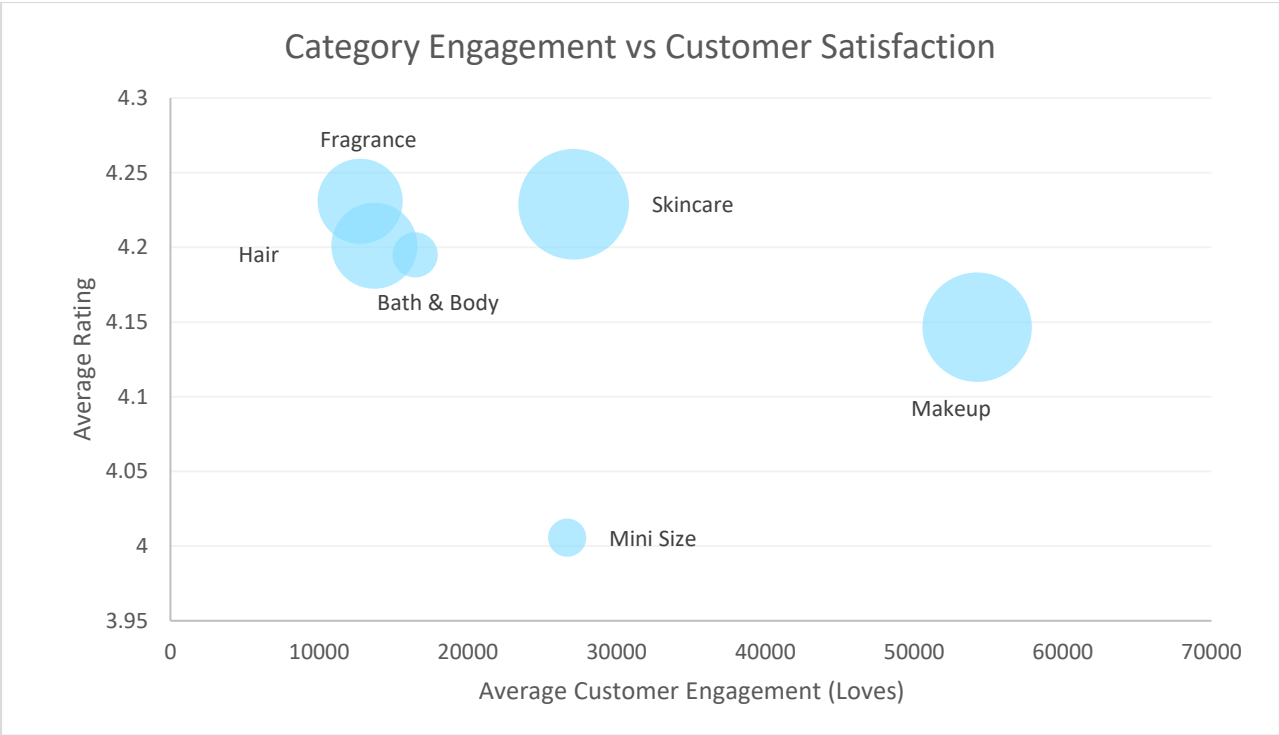
4. Category Roles Are Structurally Different

Distinct categories play different strategic roles within the portfolio:

- Makeup acts as a primary engagement engine, generating the highest customer attention
- Skincare achieves stronger average ratings but lower engagement intensity
- Mini-size products show high curiosity but weaker satisfaction

This indicates a structural split between categories that drive traffic and those that deliver stronger customer experience.

Implication: Growth opportunities may exist in increasing visibility for high-satisfaction categories.



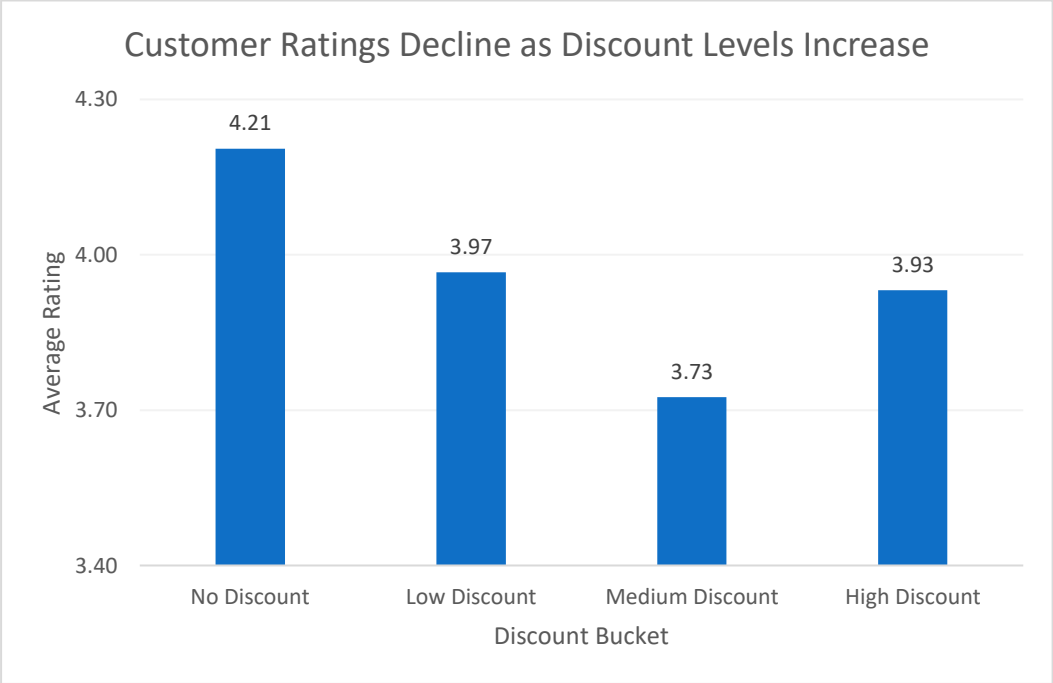
5. Discounts Do Not Drive Performance

Discounted products consistently underperform compared to non-discounted products

- Lower engagement levels
- Lower average ratings
- Premium products largely avoid discounting

Rather than boosting demand, discounts appear associated with weaker-performing products.

Implication: Discounting functions more as a corrective tool than as a reliable growth lever.



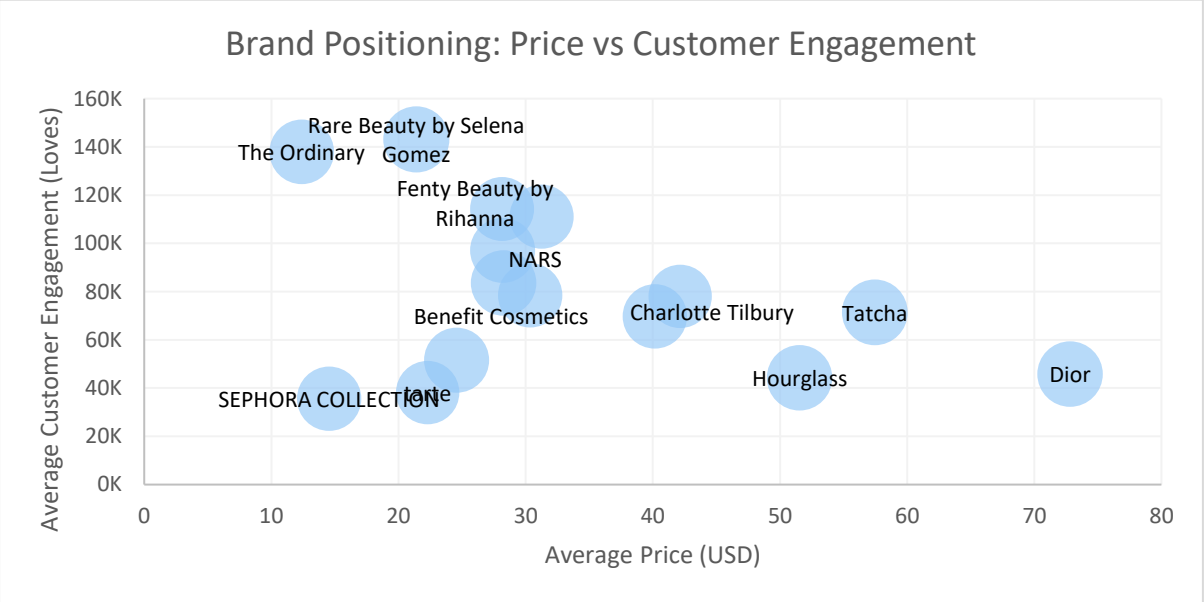
6. Brand Identity Outweighs Price Positioning

Several affordable brands generate engagement levels comparable to or higher than premium brands

- Strong value-oriented brands achieve high engagement at accessible prices.
- Premium brands maintain strong ratings but operate at smaller engagement scale.

This demonstrates that brand strength and perceived value are stronger drivers of customer interaction than price level alone.

Implication: Brand positioning and customer connection matter more than premium pricing strategy



7. Marketing Labels Have Limited Impact

Most merchandising attributes (e.g., new, limited edition, online only) show weak or negative relationships with engagement.

- New products receive higher ratings but lower engagement, reflecting lifecycle effects.
- Limited edition and online only products underperform in both engagement and ratings
- Platform exclusivity is the only attribute showing a consistent positive signal

Implication: Product performance is driven more by core value perception than by marketing labels.

Overall Finding

Across categories, pricing tiers, and brands, the portfolio demonstrates a consistent pattern:

Customer attention s driven by accessibility and brand appeal, while satisfaction follows a separate quality-driven dynamic.

This structural mismatch between engagement and satisfaction becomes the foundation for the strategic insights and recommendations that follow.

Strategic Insights

This section moves beyond individual findings to explain what the portfolio structure actually means from a strategic perspective. When viewed together, the results reveal a clear pattern in how customer attention, value perception, and brand positioning interact.

1. **Engagement Is Driven by Accessibility, Not Premium Positioning**
   Across categories and brands, customer engagement is heavily concentrated in budget-priced products. While premium products maintain slightly higher ratings, they do not attract comparable interaction levels.

   This indicates that:

- Customer engage primarily with products that feel accessible
- Premium pricing plays a positioning role rather than a traffic-driving role

Strategic Implication: Portfolio growth depends more on perceived value and accessibility than on premium price tiers.

2. **Engagement and Satisfaction Represent Two Different Success Models**
   The analysis consistently shows that highly engaged products are not always the highest rated.
   Two distinct performance patterns emerge:

- Visibility-driven success: products with strong engagement driven by brand pull, trend relevance, or exposure.
- Quality-driven success: products with strong ratings but lower engagement intensity.

This separation suggests that engagement metrics alone may overstate product success.

Strategic Implication: Decision-making should balance attention metrics with satisfaction signals to avoid over-investing in hype-driven products.

3. **Brand Strength Outweighs Pricing Strategy**
   Several affordable brands achieve engagement levels comparable to or exceeding premium brands, despite lower average prices.

   This suggests:

- Brand identity and perceived value influence customer behavior more strongly than price level.
- Premium pricing alone is insufficient to generate engagement without strong brand connection.

Strategic implication: Strengthening brand positioning and value perception may deliver greater returns than price-based repositioning.

4. **Promotional Tactics Have Limited Structural Impact**
   Discounting and most merchandising labels show weak or negative relationships with both engagement and ratings. Platform exclusivity stands out as the only attribute associated with consistently stronger performance.

   This indicates:

- Promotions are not primary growth drivers.

- Discounts may be reactive rather than strategic.
- Structural brand and portfolio factors matter more than short-term merchandising tactics.

Strategic Implication: Long-term performance is better supported by product value and brand strength than by promotional mechanics.

5. **The Portfolio Faces Engagement Concentration Risk**
   A large share of customer interaction is concentrated in lower-priced products and specific high-visibility categories.

   While this drives scale, it creates structural dependency:

- Engagement performance becomes sensitive to shifts in value-focused segments.
- Premium and high-satisfaction areas remain under-leveraged as growth drivers.

Strategic Implication: Diversifying engagement sources across categories and price tiers could reduce long-term risk and support more balanced growth.

Strategic Takeaway

Taken together, the analysis suggests that:

> The portfolio succeeds by attracting attention through affordability and brand appeal, but long-term value creation depends on strengthening the link between engagement and genuine customer satisfaction.

This insight forms the foundation for the recommendations that follow.

Recommendations

Based on the portfolio patterns identified, the following recommendations focus on improving long-term performance by aligning customer engagement with sustainable satisfaction and stronger brand positioning.

1. **Strengthen High-Satisfaction Categories Through Visibility Expansion**
   Categories such as skincare show strong customer ratings but comparatively lower engagement. This suggests quality exists but discovery and attention are limited.

   Actions

- Increase marketing visibility for high-rated categories.
- Feature high-performing skincare products in homepage placements and campaigns.
- Use cross-category bundling (e.g., makeup → skincare journeys) to drive discovery.

   Expected Impact

- Diversifies engagement sources.
- Improves long-term customer satisfaction outcomes.
- Reduces over-reliance on attention-driven categories.

2. **Optimize the Budget Segment Without Over-Relying on Discounts**

Budget products are the primary engagement engine, but discounting does not improve performance and correlates with weaker ratings.

Actions

- Maintain competitive pricing rather than frequent discounting.
- Prioritize quality improvements and product positioning over promotions.
- Use discounts selectively for inventory management rather than growth.

Expected Impact

- Protects perceived value.
- Supports healthier brand perception.
- Improves engagement quality rather than short-term spikes.

3. **Use Engagement and Satisfaction as Dual Performance Metrics**
   High engagement does not consistently indicate strong customer experience. Relying on popularity alone risks scaling hype-driven products.

   Actions

- Introduce a balanced performance score combining engagement and ratings.
- Flag products with high engagement but low ratings for quality review.
- Prioritize investment in products that perform well across both dimensions.

Expected Impact

- Improves decision quality in merchandising and promotion.
- Reduces long-term customer trust risk.
- Aligns visibility with genuine product performance.

4. **Scale Value-Driven Brands With Strong Engagement Efficiency**
   Several affordable brands demonstrate engagement levels comparable to premium brands, indicating strong perceived value and brand connection.

   Actions

- Expand assortment and visibility for high-efficiency value brands.
- Use these brands as acquisition drivers in campaigns.
- Study messaging and positioning strategies behind their success.

Expected Impact

- Higher engagement returns without premium pricing pressure.
- Stronger customer acquisition at accessible price points.
- Improved portfolio efficiency.

5. **Treat Premium Products as Positioning Assets, Not Volume Drivers**
   Premium products receive slightly higher ratings but operate at smaller engagement scale, suggesting their role is brand positioning rather than traffic generation.

   Actions

- Focus premium strategy on storytelling, expertise, and exclusivity.

- Avoid aggressive discounting that weakens premium perception.
- Use premium products to elevate overall brand credibility.

Expected Impact

- Maintains portfolio prestige layer.
- Supports pricing architecture without forcing volume expectations.
- Creates clearer segmentation between value and premium offerings.

6. **Expand Platform-Exclusive Strategy Selectively**
   Platform exclusivity is the only merchandising attribute consistently associated with stronger performance.

   Actions

- Increase platform-exclusive launches for proven categories or brands.
- Highlight exclusivity clearly in merchandising and promotions.
- Use exclusivity to differentiate rather than rely on limited-edition tactics.

Expected Impact

- Higher engagement and perceived uniqueness.
- Stronger platform loyalty.
- Improved competitive differentiation.

Overall Recommendation Theme

The portfolio should shift from a promotion-driven mindset toward a value-and-quality-driven strategy, where:

- affordable products drive reach,
- high-satisfaction products drive retention,
- and premium offerings strengthen positioning.

Conclusion & Next Steps

Conclusion

This analysis shows that portfolio performance is driven less by pricing tactics or promotional labels and more by accessibility, brand strength, and perceived value. Customer engagement is heavily concentrated in budget-oriented products, while higher-rated categories and premium offerings play smaller, positioning-focused roles.

A clear structural pattern emerges:

- Engagement and satisfaction do not always align.
- Affordability drives attention at scale.
- Brand identity influences performance more strongly than price level alone.

- Promotional mechanisms such as discounting have limited impact on sustained performance.

Together, these findings suggest that long-term growth will come not from increasing promotions or pricing complexity, but from better aligning visibility with products that deliver strong customer experiences.

Next Steps

To translate insights into measurable business outcomes, the following next steps are recommended:

1. **Establish a Balanced Performance Framework**
   Track products using both engagement and satisfaction metrics to avoid over-prioritizing hype-driven performance.

2. **Pilot Visibility Experiments**
   Increase exposure for high-rated but lower-engagement categories (e.g., skincare) and monitor engagement lift.

3. **Refine Portfolio Strategy by Role**
   Define clear roles for budget, premium, and category segments- traffic drivers, satisfaction builders, and positioning assets.

4. **Review Discounting Strategy**
   Shift discount usage toward inventory management rather than demand generation.

5. **Deepen Brand-Level Analysis**
   Investigate high-performing value brands to identify replicable drivers of engagement and perceived value.

Closing Statement

The portfolio demonstrates strong customer attention, but future growth depends on converting that attention into sustainable satisfaction. By balancing accessibility with quality-led visibility, the business can reduce structural dependency on value-driven engagement and build a more resilient, long-term growth model.